



Department of Mathematics and Natural Sciences
Final Examination
Fall 2023
Course Title: Elements of Statistics and Probability (STA 201)
Section: 32
Total Marks: 35

Time: 1 hour and 30 minutes

Date: December 17, 2023

Special Note: Answer all questions

1. Problem: A grocery shop has the following figures of sales for the past year of 8 sales persons, who have varying years of experience are given in the following table: **[5.0+2.0+1.0+2.0=10.0]**

Sales Person	1	2	3	4	5	6	7	8
Years of Experience	2	3	4	6	7	8	10	11
Annual Sales (Thousands)	3*	4*	5*	6*	7*	8*	9*	10*

Here, * = Last digit of your ID

* = Second last digit of your ID

- Using simple linear regression, find the equation of the regression line to predict annual sales based on the years of experience.
- Interpret the intercept and the slope of the regression coefficients equation in the context of this scenario.
- Predict the annual sales for the years of experience of 15.
- Suppose pearson correlation coefficient (r) for the above scenario is 0.93. Find the coefficient of determination and interpret your result.

$$\hat{\beta} = \frac{\sum_1^n x_i y_i - n\bar{x}\bar{y}}{\sum_1^n x_i^2 - n\bar{x}^2}$$
$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$$

2. In a store, there are two brands of laptops: Brand X and Brand Y. Brand X has a market share of 70%, and 5% of its laptops are reported to have technical issues. Brand Y has the remaining market share, and 10% of its laptops are reported to have technical issues. If a customer reports a technical issue with their laptop, what is the probability that it is not from a Brand Y laptop? **[3.0]**
3. A jar contains 8 red balls and 12 blue balls. Three balls are drawn without replacement. If the first two balls drawn are red and blue successively, what is the probability that the third ball is blue? **[2.0]**
4. X is a discrete random variable having the following probability mass function:

$X = x$	1	2	3	4	5	6	7
$P(X = x)$	k	2k	2k	3k	k^2	$2k^2$	$7k^2 + k$

Determine the constant k

[Hints: K must be positive]

[3.0]

5. $f(x, y) = 2k(x + y) \quad 0 < x < 1, 0 < y < 1$

- i. Find the value of k [3.0+4.0=7.0]
- ii. Calculate $E(Y|x = 0.3)$

6. Suppose a biased coin has a probability of success (landing heads) on any given trial equal to 0.3. What is the probability that the first success occurs at least after failing two times? [3.0]

7. Suppose the time (in minutes) it takes for a computer system to fail follows an exponential distribution with a rate parameter $\lambda=0.01$. Find the probability that the system will fail within the first 60 minutes. [2.0]

8. If X is a normal random variable with parameters $\mu = 2$ and $\sigma = 3$ [2.0+3.0=5.0]

Determine,

- i. $P(2 < X < 5)$
- ii. $P\{|X - 3| > 6\}$

$$[\Phi(2.33) = 0.9901, \Phi(1) = 0.8413, \Phi(0) = 0.5, \Phi(-1.67) = 0.0475]$$